Data-Driven Identification and MPC of Coupled Tank System using Bilinear Koopman Realizations and Physics-Informed Neural Networks

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***Abstract*— Real-time control of multivariable nonlinear processes requires models balancing high fidelity with computational tractability. This paper compares two data-driven paradigms: Bilinear Koopman Realizations and Physics-Informed Neural Networks (PINNs). While standard Koopman approaches seek global linearization, we leverage a bilinear framework in the lifted functional space to preserve the natural coupling of control-affine systems. Simultaneously, PINNs ensure physical consistency by embedding conservation laws into the learning objective. To facilitate high-performance control, both surrogate models are integrated into a Nonlinear Model Predictive Control (NMPC) scheme using the CasADi framework, enabling efficient gradient-based optimization. Simulation results for a quadruple tank system demonstrate that the Bilinear Koopman model offers superior tracking accuracy and robustness compared to the PINN-based approach. We conclude that the Bilinear Koopman realization, by avoiding the overfitting tendencies observed in neural architectures, provides a more reliable, structure-aware solution for the identification and control of complex hydraulic benchmarks.**

I. INTRODUCTION

Real-time control of complex nonlinear processes requires models that balance accuracy with computational efficiency [1]. Traditional first-principles modeling often incurs high computational costs in online estimation and Model Predictive Control due to the recursive solution of differential equations [1]- [5]. To address these limitations, data-driven frameworks have emerged as a flexible alternative, capturing complex nonlinearities without relying on local linearization [3], [7]. Modern approaches, such as Bilinear Koopman Realizations and Physics-Informed Neural Networks, offer a scalable path for system identification by leveraging frameworks adaptable to increasing system dimensions. These methodologies provide a high-fidelity alternative that bypasses the extensive analytical redesign typically required by classical control techniques as systems become more complex [4], [7].

Within this context, the Koopman Operator provides a mathematical foundation for analyzing nonlinear dynamics by acting as an infinite-dimensional linear operator in the observables space [3], [6]. While traditional Extended Dynamic Mode Decomposition (EDMD) seeks linear lifting, recent advances emphasize Bilinear Koopman Realizations. As argued in [16], every control-affine system admits a bilinear realization in the lifted space, even when finite-dimensional linear representations fail. This framework enables structured control while preserving natural state-input coupling [2], [3]. However, despite their algebraic elegance, these purely data-driven approximations can struggle without enforced physical consistency, particularly under measurement uncertainty.

To mitigate these generalization and noise sensitivity issues, Physics-Informed Neural Networks (PINNs) integrate differential equations as constraints into the loss function [8]. This architecture enables accurate system identification even with sparse datasets, as the model is forced to adhere to governing physical principles [4], [12]. The resulting synergy between deep learning and physics-based regularization enhances robustness against noise, overcoming the limitations of traditional black-box machine learning while maintaining the flexibility of data-driven methods [4], [13].

The comparative advantages of structure-aware bilinear operators versus physics-regularized learning set the stage for their application in demanding control benchmarks [10]-[12]. In this article, we perform a comparative analysis of Bilinear Koopman and PINN-based modeling for the identification and predictive control of the nonlinear quadruple tank system originally proposed in [9]. This multivariable benchmark is characterized by complex cross-coupled dynamics and square-root nonlinearities [2], [9]. We evaluate both approaches using simulation-generated data to assess their performance and prediction accuracy in a Nonlinear Model Predictive Control framework. The remainder of the paper is organized as follows: Section II defines the problem and the models. Section III outlines the implementation. Section IV discusses simulation results and Section V provides concluding remarks.

II. PRELIMINARIES

This section formalizes MPC optimization, Koopman Operator, Bilinear Koopman Realizations and PINN architectures, establishing the mathematical foundation for system identification and control.

A. Model Predictive Control

Model Predictive Control (MPC) computes optimal control actions by solving a constrained optimization problem over a finite time horizon at each sampling instant k [5]. Unlike

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traditional control laws, MPC leverages a surrogate model to predict the future evolution of the system, allowing the controller to anticipate complex dynamics and counteract disturbances before they affect the process.

The framework guides the system by minimizing a cost functional J over a prediction horizon N, which balances tracking performance, actuator effort, and smoothness:

$$\min_{\mathbf{u}_0,\ldots,\mathbf{u}_{N-1}} J = \sum_{i=0}^{N-1} \left(\|\hat{\mathbf{h}}_{k+i+1} - \mathbf{r}_{k+i+1}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_i\|_{\mathbf{R}}^2 + \|\Delta\mathbf{u}_i\|_{\mathbf{R}_d}^2 \right) \tag{1}$$

This optimization is subject to the system dynamics and physical operational limits:

$$\begin{aligned} \hat{\mathbf{x}}_{i+1} &= f_{\text{model}}(\hat{\mathbf{x}}_i, \mathbf{u}_i), \quad \hat{\mathbf{x}}_0 = \mathbf{x}_k \\ \mathbf{u}_{\min} &\le \mathbf{u}_i \le \mathbf{u}_{\max} \\ \Delta\mathbf{u}_{\min} &\le \mathbf{u}_i - \mathbf{u}_{i-1} \le \Delta\mathbf{u}_{\max} \end{aligned}$$

In this formulation, $\hat{\mathbf{h}}$ represents the predicted tank levels and $\mathbf{r}$ is the desired reference. The matrix $\mathbf{Q}$ defines the importance of tracking precision, $\mathbf{R}$ represents the penalty on energy consumption or control effort, and $\mathbf{R}_d$ accounts for the desire for temporal smoothness in the control signal. Since the controller's success depends entirely on the fidelity of the mapping f_{model}, this study evaluates the integration of Bilinear Koopman Realizations and Physics-Informed Neural Networks as the predictive engines within this optimization framework.

B. Koopman Operator Theory

Koopman operator theory provides a mathematical foundation for analyzing nonlinear dynamics by shifting the perspective from state-space trajectories to the evolution of functions. It describes nonlinear systems through an infinite-dimensional linear operator $\mathcal{K}$ that acts on a space of scalar-valued observables g, such that $\mathcal{K}g = g \circ \mathbf{f}$ [6]. In practice, this framework seeks to find a finite-dimensional *lifting* mapping $\mathbf{z} = \psi(\mathbf{h})$ that embeds the nonlinear manifold into a higher-dimensional space where the dynamics appear linear. However, while this linear lifting is mathematically elegant for autonomous systems, the introduction of external control inputs often breaks the global linearity, necessitating a more flexible representation to capture state-input interactions.

C. Bilinear Koopman Realizations

To bridge the gap between global linearization and control-affine systems, recent advances favor Bilinear Koopman Realizations over traditional linear approximations. As established in [16], every control-affine system admits a bilinear representation in the lifted functional space, even when a strictly linear one is unattainable. This approach approximates the dynamics by allowing the control input $\mathbf{u}$ to act multiplicatively with the lifted states $\mathbf{z}$:

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \sum_{j=1}^{m} (\mathbf{B}_j \mathbf{z}_k) u_{j,k} + \mathbf{C}\mathbf{u}_k \tag{3}$$

where $\mathbf{A}, \mathbf{B}_j$, and $\mathbf{C}$ capture autonomous, bilinear, and direct control effects, respectively. This structure retains the computational benefits of structured models while significantly improving prediction fidelity. Nevertheless, the accuracy of these bilinear models remains heavily dependent on the manual selection of basis functions and the quality of the dataset. This limitation—the potential lack of physical consistency in purely data-driven mappings—motivates the integration of physics-based constraints through neural architectures.

D. Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) approximate dynamical system solutions by embedding governing physical equations directly into the network's optimization objective. The framework trains a neural network $\hat{\mathbf{h}}(t) = \mathcal{N}(t, \mathbf{v}; \boldsymbol{\theta})$ to represent system trajectories while strictly adhering to physical laws [4], [8].

A composite loss function $\mathcal{L}(\boldsymbol{\theta})$ governs the training process:

$$\mathcal{L}(\boldsymbol{\theta}) = w_d \mathcal{L}_{data} + w_p \mathcal{L}_{phys} + w_{ic} \mathcal{L}_{ic},$$

where weighting coefficients w_i balance each term's contribution. Specifically, $\mathcal{L}_{data}$ quantifies the mean squared error against measured data, while $\mathcal{L}_{ic}$ enforces initial conditions. To maintain physical consistency, the residual term $\mathcal{L}_{phys}$ leverages automatic differentiation to evaluate the governing law:

$$\mathbf{r}(t) = \frac{d}{dt}\hat{\mathbf{h}}(t) - \mathbf{f}(\hat{\mathbf{h}}(t), \mathbf{v}(t); \boldsymbol{\omega}).$$

By minimizing the norm of this residual over collocation points, the network converges toward a solution that satisfies fundamental physics principles. This approach effectively bridges the gap between purely data-driven methods and first-principles modeling.

III. PROBLEM FORMULATION

This section establishes the mathematical foundation of the quadruple tank system and formalizes the data-driven identification and control objectives addressed in this study.

A. Quadruple Tank System Dynamics

The quadruple-tank process serves as a multivariable (MIMO) benchmark for control, characterized by complex cross-coupled dynamics. As Fig. 1 illustrates, the setup utilizes two pumps to distribute flow among four interconnected tanks through a specific valve configuration.

Mass conservation and Bernoulli's principle define the following nonlinear state-space representation:

$$\begin{aligned} \dot{h}_1 &= -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1}\sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1} v_1, \\ \dot{h}_2 &= -\frac{a_2}{A_2}\sqrt{2gh_2} + \frac{a_4}{A_2}\sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2} v_2, \\ \dot{h}_3 &= -\frac{a_3}{A_3}\sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3} v_2, \\ \dot{h}_4 &= -\frac{a_4}{A_4}\sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4} v_1, \end{aligned}$$

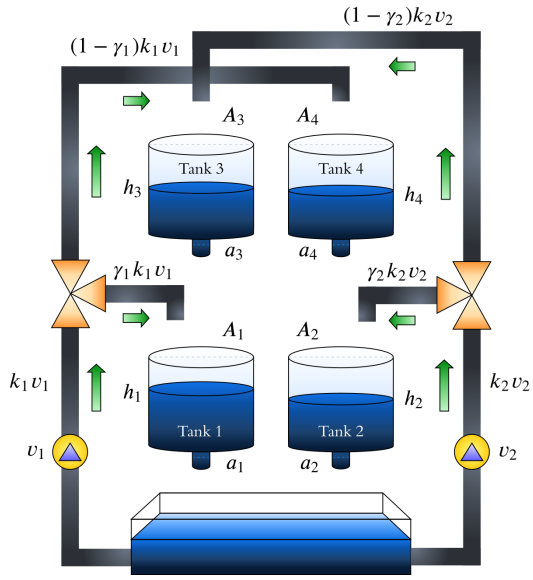

Fig. 1. Quadruple Tank System Representation.

where h_i represents the water level in each tank and v_i denotes the pump voltages. The parameters A_i and a_i correspond to the cross-sectional and outlet areas, respectively; k_i are the flow constants, and $\gamma_1, \gamma_2 \in [0,1]$ represent the valve settings that determine the flow distribution. The system output vector $\mathbf{y}$ integrates the measured levels of all four tanks:

$$\mathbf{y} = [h_1, h_2, h_3, h_4]^\top.$$

Torricelli's Law introduces square-root nonlinearities, while the system equations reveal strong cross-coupling; specifically, the levels h_3 and h_4 directly drive the dynamics of the lower tanks. These structural interdependencies demand high-fidelity identification to achieve effective regulation within the control framework [9].

B. Identification and Control Objectives

This work primarily aims to identify accurate surrogate models for the quadruple tank system using two distinct data-driven paradigms: Bilinear Koopman Realizations and Physics-Informed Neural Networks (PINNs). These models must capture nonlinear interactions and cross-coupling effects to enable precise trajectory tracking. Consequently, the control problem involves designing an MPC strategy to regulate the lower tank levels (h_1, h_2) by manipulating pump voltages $\mathbf{v} = [v_1, v_2]^\top$.

IV. METHODOLOGY

This section details the data-driven identification strategy and the MPC framework designed to regulate the quadruple tank's nonlinear cross-coupled dynamics.

A. Bilinear Koopman Identification

The identification process implements the transition function f_{model} by mapping tank levels $\mathbf{h} \in \mathbb{R}^4$ into a higher-dimensional feature space. To capture Bernoulli-driven nonlinearities and trigonometric couplings, we employ the dictionary $\mathbf{z} = \boldsymbol{\psi}(\mathbf{h}) = [\mathbf{h}, \sqrt{|\mathbf{h}|}, \sin(\mathbf{h}), \cos(\mathbf{h})]^\top$.

To identify the matrices $(\mathbf{A}, \mathbf{B}_j, \mathbf{C})$, we construct a regression matrix $\mathbf{X}_{\text{reg}}$ containing the lifted states, control inputs, and their bilinear interactions $\mathbf{z}_k u_{j,k}$. The operator weights $\mathbf{W} = [\mathbf{A}, \mathbf{B}_1, \ldots, \mathbf{B}_m, \mathbf{C}]$ are estimated via Ridge regression:

$$\min_{\mathbf{W}} \|\mathbf{Z}_{\text{next}} - \mathbf{W}\mathbf{X}_{\text{reg}}\|_F^2 + \alpha \|\mathbf{W}\|_F^2 \tag{4}$$

where α ensures numerical stability against sensor noise. This structure allows the NMPC to predict system evolution in the lifted manifold, which is then projected back to the physical space for real-time level estimation.

B. PINN Identification

As an alternative, a PINN identifier approximates the function f_{model} through a deep residual network $\mathcal{N}$. The architecture processes the input vector $[\mathbf{x}_k, \mathbf{u}_k] \in \mathbb{R}^6$ to predict the subsequent state $\hat{\mathbf{x}}_{k+1}$. To enforce physical consistency, the training process minimizes the composite loss $\mathcal{L} = w_d \mathcal{L}_{data} + w_r \mathcal{L}_{res} + w_{ic} \mathcal{L}_{ic}$, where the residual term $\mathcal{L}_{res}$ penalizes deviations from the discrete-time mass balance:

$$\mathbf{R}_k = \hat{\mathbf{x}}_{k+1} - \left(\mathbf{x}_k + \Delta t \cdot \mathbf{f}_{phys}(\mathbf{x}_k, \mathbf{u}_k)\right).$$

To maintain numerical stability, f_{model} utilizes a scaling mapping based on the training extrema $[h_{\min}, h_{\max}]$. This normalization, in conjunction with optimization clipping, restricts the predicted trajectories $\hat{\mathbf{x}}$ to the feasible hydraulic range during MPC execution.

C. MPC Approach

The proposed control framework utilizes a Nonlinear Model Predictive Control strategy centered on the CasADi software suite for numerical optimization. By leveraging algorithmic differentiation, the controller transforms the surrogate models f_{model} into differentiable expressions, enabling the use of high-performance gradient-based solvers.

For the Bilinear Koopman model, the optimization is structured as a Quadratic Programming problem and solved via the OSQP solver, exploiting the model's structured linearity in the lifted space. Conversely, the PINN-based controller handles the inherent nonlinearities of the neural architecture through a Sequential Quadratic Programming method. Both implementations enforce real-time feasibility by using a warm-start strategy from the previous solution $\mathbf{u}^*_{k-1}$. This unified optimization approach ensures that differences in control performance stem strictly from the predictive fidelity of the underlying data-driven paradigms rather than the optimization algorithm itself.

V. CASE STUDY

This section details the implemented environment, the training protocol for neural surrogates, and the physical parameters of the quadruple tank benchmark.

A. System Parameters

Plant simulation and predictive models rely on the geometric and flow constants from the Johansson benchmark [9], summarized in Table I. These values define minimum phase characteristics and the multivariable coupling strength of the system.

TABLE I
PHYSICAL PARAMETERS OF THE QUADRUPLE TANK SYSTEM.

Symbol	Description	Value	Unit
$A_{1,3}$	Area (Tanks 1 & 3)	28.00	cm^2
$A_{2,4}$	Area (Tanks 2 & 4)	32.00	cm^2
$a_{1,3}$	Outlet area (Tanks 1 & 3)	0.071	cm^2
$a_{2,4}$	Outlet area (Tanks 2 & 4)	0.057	cm^2
k_1, k_2	Pump flow constants	3.33, 3.35	cm^3/Vs
γ_1, γ_2	Flow split ratios	0.70, 0.60	–
g	Gravity	981.00	cm/s^2

B. Data Acquisition and Excitation

To achieve persistent excitation, data collection spans 5000 s with a sampling period $dt = 0.05$ s. A 4th-order Runge-Kutta method (RK45) integrates the nonlinear plant from initial conditions $\mathbf{h}_0 = [12.4, 12.7, 1.8, 1.4]^\top$ cm.

Excitation signals $\mathbf{u} = [v_1, v_2]^\top$ consist of a Pseudo-Random Binary Sequence (PRBS) within $[0, 6]$ V, switching every 50 s. Reproducibility is ensured via a Mersenne Twister (MT19937) generator with a fixed seed of 100. Furthermore, to emulate realistic conditions, additive white Gaussian noise (AWGN) with $\sigma = 0.1$ affects both control inputs and output measurements.

C. Identification Parameters and Training

The framework implements the PINN using Python and *PyTorch*, accelerating operations via an *NVIDIA GeForce GTX 1050* GPU. Both models adopt the configurations in Table II. Before training, the system normalizes measurements to $[0, 1]$ to ensure stability and faster convergence. For the PINN, the training utilizes a dual-phase strategy: an Adam optimizer performs the global search, followed by L-BFGS for second-order refinement of physical residuals.

TABLE II
PARAMETERS FOR KOOPMAN AND PINN IDENTIFICATION

Koopman Parameter	Value	PINN Parameter	Value
Observables (n_{obs})	16	Hidden Layers	4
Regularization (λ)	0.01	Neurons per Layer	48
Snapshot count (N)	5000	Activation Function	SiLU
Time step (Δt)	0.05 s	Adam Epochs	4000
$\mathbf{K}$ matrix size	4×16	L-BFGS Max Iter.	4000

D. Identification Results

Multi-step prediction performance evaluation compares Bilinear Koopman Realization and PINN trajectories against actual tank levels. Fig. 2 illustrates this comparison over a validation window (2500–5000 s). The Bilinear Koopman model achieves high structural fidelity, capturing nonlinearities without overfitting to sensor noise. While the PINN demonstrates exceptional fitting, it exhibits higher sensitivity to measurement uncertainty during recursive predictions.

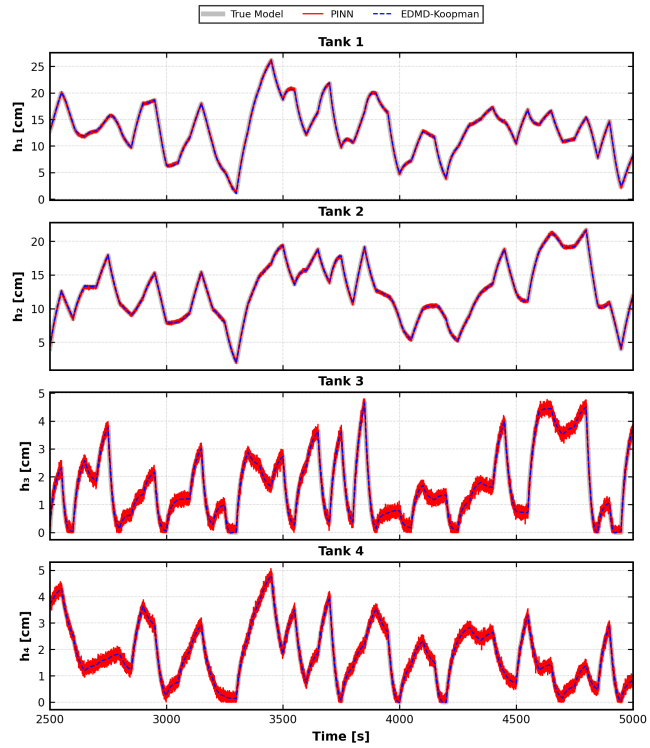

Fig. 2. Identification results: EDMD-Koopman vs PINN.

E. Model Validation Metrics

Quantitative assessment over the validation dataset reveals that the Bilinear Koopman architecture consistently outperforms the PINN approach across all state variables. As summarized in Table III, the bilinear model achieves a Variance After Filtering (VAF) above 99.9% for all tanks, with significantly lower RMSE and MAE values. This exceptional fidelity stems from the structured lifting of states, which captures the hydraulic couplings without the overfitting tendencies often found in neural architectures.

Conversely, while the PINN maintains high accuracy, its performance slightly degrades in the upper stages (h_3, h_4), with VAF values falling to 98.81% and 98.50%, respectively. This gap suggests that the Bilinear Koopman realization, by exploiting the system's natural control-affine form, provides a more robust and precise representation of the quadruple tank dynamics than the physics-regularized neural network under the evaluated conditions.

F. Controller Real-Time Execution and Feedback Loop

The NMPC operates in a closed-loop with a sampling period of $\Delta t = 1$ s. At each step, the controller receives filtered measurements and computes the optimal sequence using a warm-start strategy from $\mathbf{u}^*_{k-1}$. While the Bilinear Koopman model enables near-instantaneous QP solutions,

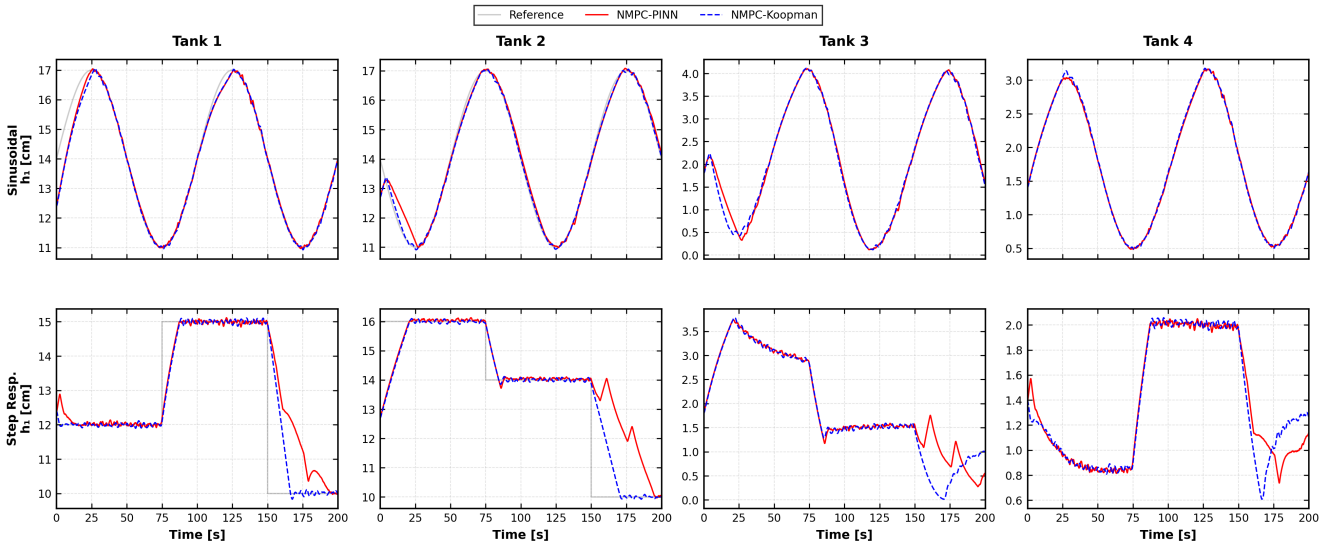

Fig. 3. Control performance comparison between MPC-Koopman and MPC-PINN: Sinusoidal Tracking (left) and Step Response (right).

TABLE III

DETAILED PREDICTION METRICS PER TANK.

Model	Tank	RMSE (cm)	MAE (cm)	VAF (%)
Bilinear Koopman	h_1	**0.0907**	**0.0710**	**99.97**
	h_2	**0.0665**	**0.0534**	**99.98**
	h_3	**0.0131**	**0.0101**	**99.99**
	h_4	**0.0090**	**0.0068**	**99.99**
PINN	h_1	0.1419	0.1132	99.93
	h_2	0.1411	0.1127	99.89
	h_3	0.1384	0.1096	98.81
	h_4	0.1397	0.1114	98.50

the PINN-based approach maintains a constant execution frequency via CasADi. Robustness is verified by injecting AWGN into sensors ($\sigma = 0.1$) and actuators ($\sigma = 0.15$), evaluating resilience to stochastic disturbances.

Table IV summarizes the NMPC hyperparameters used for both predictive models. The weighting matrices prioritize tracking accuracy in the lower tanks ($\mathbf{Q}$) while ensuring control effort economy ($\mathbf{R}$) and actuator smoothness ($\mathbf{R}_d$).

TABLE IV

NMPC HYPERPARAMETERS.

Parameter	Value	Description
N	12	Prediction horizon
$\Delta\mathbf{u}_{\max}$	0.2 V / 0.5 V	Slew-rate limit
Weighting Matrices		
$q_{1,2}$	2500	Tracking weights (h_1, h_2)
$q_{3,4}$	0	Stability weights (h_3, h_4)
$\mathbf{R}$	$0.01 \cdot \mathbf{I}$	Control effort weight
$\mathbf{R}_d$	$1.0 \cdot \mathbf{I}$	Slew-rate (smoothness) weight

G. Control Scenarios and Tracking Performance

Closed-loop simulations run for 200 s ($dt = 0.05$ s) starting from a quiescent state. To evaluate agility and robustness, a Step Response Cascade first switches h_1 between 12, 15, and 10 cm every 75 s (with inverse values for h_2) to quantify transient handling. Subsequently, a MIMO Sinusoidal Tracking scenario employs the antiphase trajectory $\mathbf{r}(t) = [14, 14]^\top + [3, -3]^\top \sin(2\pi t/100)$, stressing multivariable couplings to assess the precision of the data-driven predictors.

H. Control Performance Analysis

Fig. 3 compares the closed-loop tracking performance of both architectures. While both Bilinear Koopman and PINN-based NMPC maintain high fidelity to reference trajectories despite multivariable coupling, significant differences emerge in transient behavior. Notably, the Bilinear Koopman NMPC exhibits a faster convergence and settling time compared to its PINN-based counterpart. This enhanced stability stems from the Koopman model's ability to capture the underlying system structure without incorporating the high-frequency sensor noise that the PINN architecture tended to learn during training. Consequently, the Koopman realization ensures a smoother and more efficient transition within the $[0, 6]$ V control manifold, preventing the oscillatory artifacts observed in the more sensitive neural network approach.

I. Control Validation Metrics

Table V summarizes the global error analysis for both evaluated scenarios. In sinusoidal tracking, the Bilinear Koopman NMPC achieves a superior RMSE of 0.2853 cm, surpassing the PINN-based approach. While both models perform competitively under smooth trajectories, the Koopman realization demonstrates a more precise alignment with the reference by leveraging its structured bilinear form.

The performance gap becomes more pronounced in the multistep response scenario. The Bilinear Koopman model yields an RMSE of 0.9470 cm and significantly lower energy error ($SSE = 7174.84\,\text{cm}^2$), compared to the $13011.63\,\text{cm}^2$ produced by the PINN. This difference confirms that the

TABLE V
GLOBAL PERFORMANCE METRICS FOR MPC COMPARISON.

Scenario	Metric	Bilinear Koopman	PINN	Unit
Sinusoidal Tracking	RMSE	**0.2853**	0.3003	cm
	SSE	**651.07**	721.64	cm^2
	NRMSE	**4.75**	5.01	%
Multistep Response	RMSE	**0.9470**	1.2753	cm
	SSE	**7174.84**	13011.63	cm^2
	NRMSE	**15.78**	21.26	%

neural architecture, by overfitting to high-frequency sensor noise, introduces oscillatory transients that degrade control precision. In contrast, the Koopman-based controller effectively rejects these disturbances, maintaining superior stability and tracking accuracy throughout the operational range.

VI. CONCLUSION

This study evaluated Bilinear Koopman and PINN architectures for the predictive control of a nonlinear quadruple tank system. Results demonstrate that the Bilinear Koopman model achieves superior identification and control performance by exploiting the system's control-affine structure. Unlike the PINN, which exhibited sensitivity to sensor noise during training, the Koopman framework effectively filtered stochastic disturbances, achieving a VAF above 99.9% and significantly faster stabilization in closed-loop scenarios.

In the NMPC stage, the Koopman-based controller consistently outperformed the PINN approach in both sinusoidal and multistep tracking, yielding an RMSE of 0.9470 cm and a 44.8% reduction in global SSE. The integration with CasADi enabled efficient real-time execution, where the bilinear structure favored rapid QP solutions via OSQP, while the PINN required more intensive SQP optimization. While PINNs offer high fitting potential, the structural robustness of Bilinear Koopman realizations provides a more reliable foundation for the predictive control of hydraulic processes. Future work will explore the adaptive scaling of lifting functions to maintain this fidelity under varying operational regimes.

DATA AND CODE AVAILABILITY

The source code, simulation results, and models used in this study are available to support reproducibility. These resources include the implementation of the PINN, Bilinear Koopman Realization, and NMPC architectures. Access the repository at: `https://github.com/azvcud/CoDiT_Comparison-of-PINN-and-EDMD-Koopman`.

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